

Sample and Hold (S & H)

Storing analog signals is a task frequently encountered in music synthesizers—for instance, consider the control voltage "memories" for VCOs and velocity-sensitive CV. The sample-and-hold module generates control voltage sequences that are not tied to any regular pattern.

Analog sample-and-hold circuits (hereinafter referred to simply as S&H) are no mystery to the builder of a Formant synthesizer. Any owner of an (analog) keyboard or touch-sensitive controller will appreciate their operation, which is as free as possible from "drift" and "tracking" errors. However, the S&H circuits used for the keyboard electronics are too slow to generate staircase voltages or random patterns.

Figure 1 shows the block diagram of the S&H module. The "Sample" branch consists essentially of a (triggerable) electronic switch "S", a storage capacitor C_x , and a voltage follower. Switch "S" is controlled by the comparator, whose "Sample" threshold is adjustable between 0 V and +2.5 V. A trigger threshold of +2.0 V, for example, ensures that triggering only activates at levels above +2.0 V. An LED indicator makes the "Sample" mode visible.

In the search for a suitable circuit, the choice fell on an application based on the well-known CA3080 (Figure 3). The OTA (Operational Transconductance Amplifier) replaces switch "S" from the block diagram. When a signal is present at the sample input, the output (with the switch closed) attempts to track the input voltage as rapidly as possible. Depending on the "setting" of the signal slew rate,

$$\text{max. slew rate} = \frac{I_{ABC}}{C_x} \left[\frac{V}{s} \right]$$

the switch is closed (sample mode) or opened (hold mode). Since the charging behavior is determined by the capacitance of C_2 , the signal slew rate can only be influenced by varying I_{ABC} . In sample mode, I_{ABC} is very high (the maximum slew rate is 3 V/ms), meaning that relatively short sampling times can be used.

In hold mode, the input current drops to near zero, and the voltage last present at the input is stored until the next "sample."

Following the actual S&H stage, IC2—configured as a MOSFET voltage follower—buffers the stored voltage with low output impedance. The electronic switch, implemented using OTA IC1, is controlled by comparator IC3. As soon as the trigger signal exceeds the threshold (adjustable via P1), a positive pulse—limited to approximately +3.5 V by the voltage divider (R7/P2)—is fed via R1 to the emitter of switching transistor T1. Diode D3 clamps negative signals at the comparator output to ground. LED D4 provides a visual indication of the positive pulses ("samples"), thereby facilitating precise adjustment of the switching threshold.

The operating principle of the S&H circuit is illustrated by the diagrams in Figure 2, which show the voltage waveforms over time in the various sections of the circuit.

Assembly

Assembling the circuit based on the PCB layout and component placement diagram in Figure 4 should present no difficulties, provided that standard quality criteria are observed when selecting components. Both Mini-DIP and "metal-can" types are suitable for IC1 and IC2. With the latter, the individual pins must be bent to match the pinout; however, metal-can versions (TO-5) with DIL-compatible pin arrangements are also available under the designations CA3080S and CA3080AS. It is advisable to use an IC socket for IC2 unless you possess a soldering iron with a fully potential-free tip. A film capacitor with exceptionally low leakage current is advantageous for C2. For the other capacitors, too, it does no harm to spend a little extra.

Figure 5 shows the proposed front panel layout for the sample-and-hold module. The controls are positioned alongside the input and output jacks on a "small" FORMANT front panel.

Calibrating the S&H module is remarkably simple, as there is only a single adjustment point to consider. To calibrate, connect an oscilloscope or multimeter to the wiper of P2 and turn the wiper of P1 toward ground. Ideally, apply a square wave from an LFO to the trigger input and adjust the wiper of P2 to approximately +3.5 V while the square wave is at its "upper plateau" (observe the LED indicator).

Applications

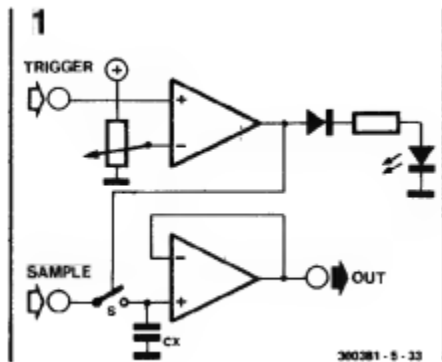
The primary applications of the S&H module likely involve generating a wide variety of staircase voltage configurations (hence the name "staircase generator") and random patterns. A particularly fascinating aspect is that this allows the synthesizer to "play" on its own; with careful adjustment of the individual parameters, one can create musically appealing sequences and modulation patterns—some of which are reminiscent of sequencers. The S&H can also be used to impart a "random" character to a sequencer signal sequence.

Literature:

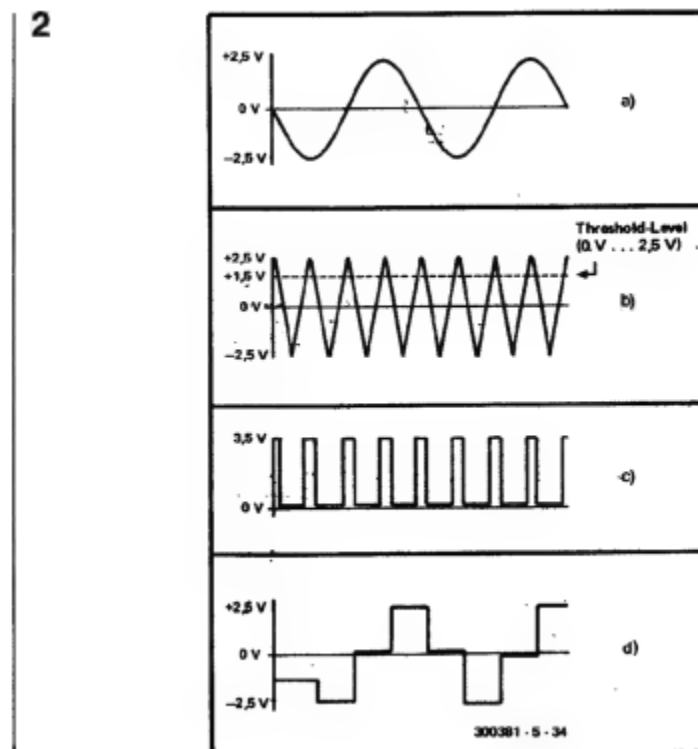
- Linear Integrated Circuits CA 3080/A/AS/E/S, RCA Data Sheets No. 475, 1974
- G.D. Shaw: P.E. Sound Synthesizer 5, Sample and Hold, Practical Electronics, June 1973, p. 506 et seq.
- H.A. Wittlinger: Application of the Operational Transconductance Amplifiers CA 3080 and CA 3080A, RCA Application Note ICAN-6668, 1973
- Sample & Hold for Synthesizers, Halbleiterheft 1980, Elektor-Verlag.

Figures

- Figure 1. Block diagram of the sample-and-hold module.

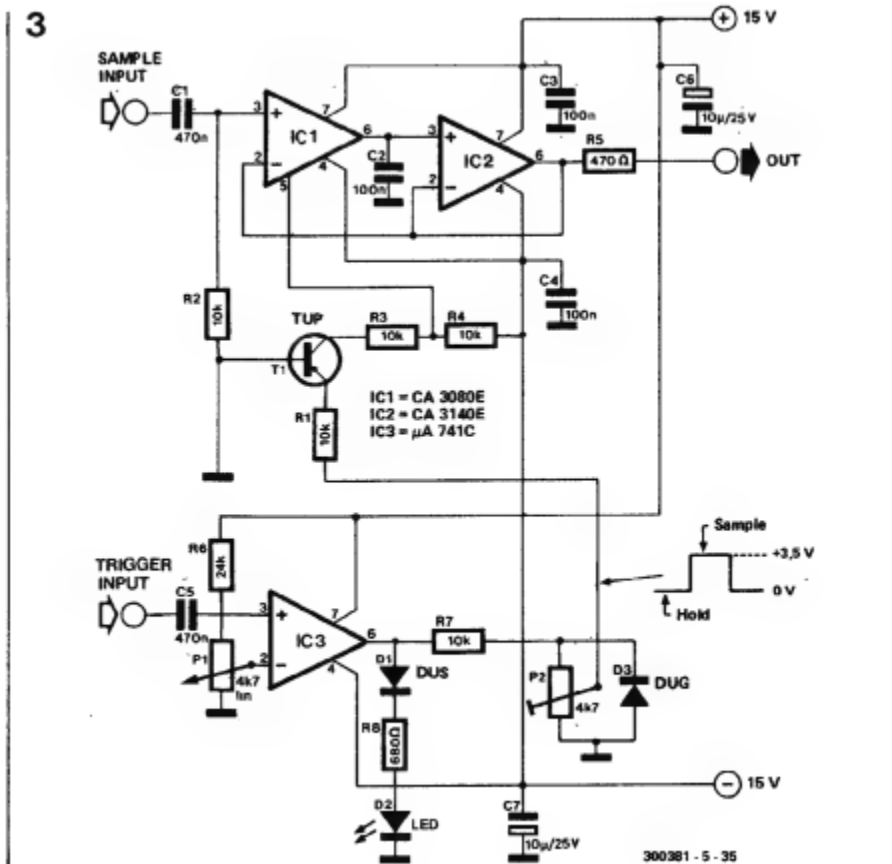


- Figure 2. The operating principle of the sample-and-hold module is easily understood from the waveform shown:

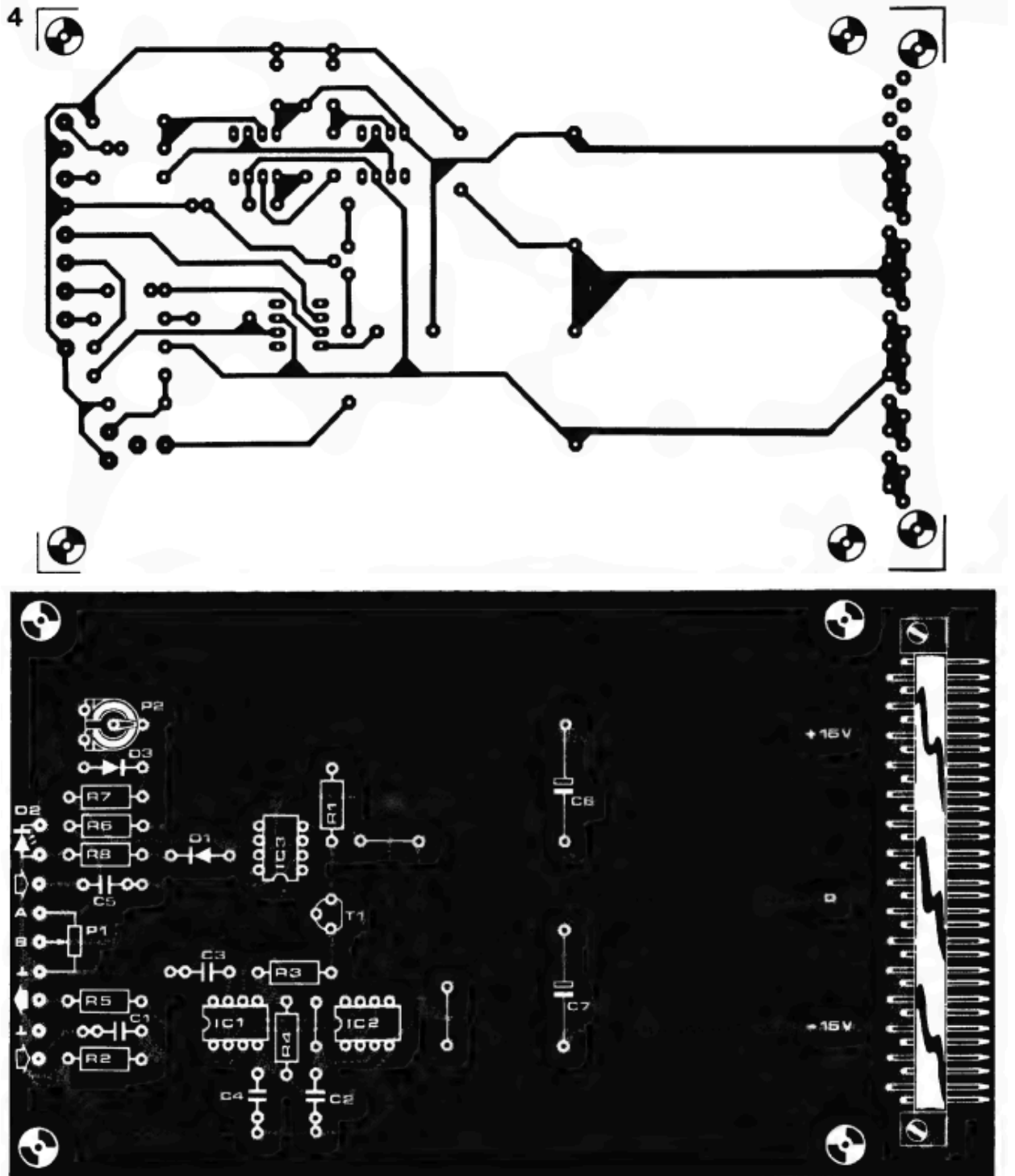


- a) Input signal (sample)
- b) Input signal (trigger)
- c) Trigger signal
- d) Output signal

- Figure 3. Complete sample-and-hold circuit for the FORMANT.



- Figure 4. Layout, component placement plan, and parts list for the S&H module.



Sample-and-Hold Bill of Materials

Resistors (carbon film, 5%):

R1, R2, R3, R4, R7 = 10 k Ω

R5 = 470 Ω

R6 = 24 k Ω (22 k Ω + 2.2 k Ω)

R8 = 680 Ω

Potentiometers:

P1 = 4.7 k Ω (linear)

P2 = 4.7 k Ω (trimmer)

Capacitors

(Film Capacitors MKH, MKS):

C1, C5 = 470 nF

C2, C3, C4 = 100 nF

C6, C7 = 10 μ F/25 V

Semiconductors:

D1 = DUS (signal diode)

D2 = LED

D3 = DUG (signal diode)

T1 = TUP (general-purpose PNP transistor)

IC1 = CA3080E (Mini-DIP)

IC2 = CA3140E (Mini-DIP)

IC3 = μ A741C (Mini-DIP)

Miscellaneous:

3x 3.5 mm jack sockets

1x Rotary knob for 6 mm shaft

1x 31-pin pin header or solder pins